Lecture-3

$$x+y+z=3$$
$$2x+2y+5z=9$$
$$3x+6y+4z=12$$

$$\left(\begin{array}{ccc|c} \boxed{1} & 1 & 1 & 3 \\ 2 & 2 & 5 & 9 \\ 3 & 6 & 4 & 12 \end{array}\right)$$

$$\left(\begin{array}{ccc|c} 1 & 1 & 1 & 3 \\ 0 & 0 & \boxed{3} & 3 \\ 0 & 3 & 1 & 3 \end{array}\right) \quad \begin{array}{l} R_2' = R_2 - 2R_1 \\ R_3' = R_3 - 3R_1 \end{array} \sim \left(\begin{array}{ccc|c} \boxed{1} & 1 & 1 & 3 \\ 0 & \boxed{3} & 1 & 3 \\ 0 & 0 & \boxed{3} & 3 \end{array}\right) \quad R_2 \leftrightarrow R_3$$

Because we have pivots in second column at third row, $\begin{pmatrix} 1 \\ 0 \\ 3 \end{pmatrix}$ that's why we need the swap.

Nice! We get row echelon form. $3z=3 \Rightarrow \underline{z=1}$

2nd Eq: $3y+z=3 \xrightarrow{z=1} 3y+1=3 \Rightarrow y=2/3$

1st Eq: $x+y+z=3 \xrightarrow{z=1,\ y=2/3} x = 3-\frac{2}{3}-1 = \frac{9-2-3}{3} = \frac{4}{3}$

Q2

$$\left(\begin{array}{ccc|c} 0 & 1 & 1 & 2 \\ \boxed{1} & 1 & 1 & 3 \\ 2 & 3 & 1 & 5 \end{array}\right)$$

(the first pivot is at the second column but we need pivot in 1st column, $\boxed{\text{then swap}}$!)

$$\left(\begin{array}{ccc|c} \boxed{1} & 1 & 1 & 3 \\ 0 & 1 & 1 & 2 \\ 2 & 3 & 1 & 5 \end{array}\right) \sim \left(\begin{array}{ccc|c} 1 & 1 & 1 & 3 \\ 0 & \boxed{1} & 1 & 2 \\ 0 & 1 & -1 & -1 \end{array}\right) \sim \left(\begin{array}{ccc|c} 1 & 1 & 1 & 3 \\ 0 & 1 & 1 & 2 \\ 0 & 0 & -2 & -3 \end{array}\right)$$

Q3

$$\left(\begin{array}{ccc|c} \boxed{1} & 1 & 1 & 3 \\ 2 & 2 & 2 & 6 \\ 1 & 2 & 3 & 4 \end{array}\right) \sim \left(\begin{array}{ccc|c} 1 & 1 & 1 & 3 \\ 0 & 0 & 0 & 0 \\ 0 & 1 & 2 & 1 \end{array}\right) \quad \left(\begin{array}{l} R_2' = R_2 - 2R_1 \\ R_3' = R_3 - R_1 \end{array}\right)$$

$$\sim \left(\begin{array}{ccc|c} 1 & 1 & 1 & 3 \\ 0 & 1 & 2 & 1 \\ 0 & 0 & \boxed{0} & 0 \end{array}\right)$$

z

we miss the pivot here
the column related z-variable
that's why we choose z as free variable

Surviving fq^n

$x + y + z = 3$

$y + 2z = 1$

$t \in \mathbb{R}$

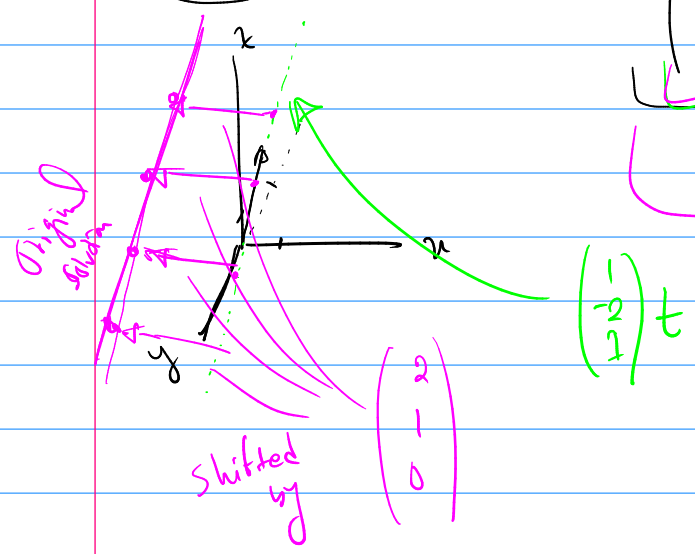
$z = t$ (arbitrary value)

$y = 1 - 2z = 1 - 2t$

$x = 3 - y - z = 3 - 1 + 2t - t$
$= 2 + t$

$$\underbrace{\begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 2+t \\ 1-2t \\ t \end{pmatrix}}_{\text{Parametric Solution}} = \begin{pmatrix} 2+t \\ 1-2t \\ 0+t \end{pmatrix}$$

$$= \underbrace{\begin{pmatrix} 2 \\ 1 \\ 0 \end{pmatrix} + \begin{pmatrix} 1 \\ -2 \\ 1 \end{pmatrix} t}_{\text{General Solution}}$$



Q4

$$\left(\begin{array}{ccc|c} \boxed{1} & 2 & -3 & -1 \\ 3 & -1 & 2 & 7 \\ 5 & 3 & -4 & 2 \end{array}\right) \to \left(\begin{array}{ccc|c} 1 & 2 & -3 & -1 \\ 0 & \boxed{-7} & 11 & 10 \\ 0 & -7 & 11 & 7 \end{array}\right)$$

$$\left(\begin{array}{ccc|c} 1 & 2 & -3 & -1 \\ 0 & -7 & 11 & 10 \\ 0 & 0 & 0 & -3 \end{array}\right) \to \underbrace{0x+0y+0z}=-3$$

$$0=-3$$

No solution.

Q5

$$\left(\begin{array}{ccc|c} 1 & 1 & 1 & 6 \\ 1 & 2 & 3 & 10 \\ 1 & 2 & \boxed{\lambda} & \boxed{\mu} \end{array}\right) \sim \left(\begin{array}{ccc|c} 1 & 1 & 1 & 6 \\ 0 & 1 & 2 & 4 \\ 0 & 1 & \lambda-1 & \mu-6 \end{array}\right)$$

$$\left(\begin{array}{ccc|c} \boxed{1} & 1 & 1 & 6 \\ 0 & \boxed{1} & 2 & 4 \\ 0 & 0 & \lambda-3 & \mu-10 \end{array}\right)$$

$az=*$ or $z=\frac{*}{a}$

there must be pivots for unique solution

For unique solution

$\lambda-3 \neq 0$

$\Rightarrow \lambda \neq 3$

Many solution: $\lambda-3=0$ & $\mu-10=0 \Rightarrow \lambda=3$ & $\mu=10$

No solution: $\lambda-3=0$ & $\mu-10\neq 0 \Rightarrow \lambda=3$ & $\mu\neq 10$

$(0\ 0\ 0 \mid *)$

Q5

$$A_{3\times 3} = \begin{pmatrix} 1 & 2 & 3 \\ 2 & 5 & 3 \\ 1 & 0 & 8 \end{pmatrix} \qquad A^{-1} = \frac{\mathrm{Adj}(A)}{\underbrace{\det(A)}}$$

$[A|I]$

$$\left(\begin{array}{ccc|ccc} 1 & 2 & 3 & 1 & 0 & 0 \\ 2 & 5 & 3 & 0 & 1 & 0 \\ 1 & 0 & 8 & 0 & 0 & 1 \end{array}\right) \sim \left(\begin{array}{ccc|ccc} 1 & 2 & 3 & 1 & 0 & 0 \\ 0 & 1 & -3 & -2 & 1 & 0 \\ 0 & -2 & 5 & -1 & 0 & 1 \end{array}\right)$$

$$\begin{array}{ccc} 0 & 1 & 0 \\ 2 & 0 & 1 \\ \hline -2 & 1 & 0 \end{array}$$

$$\left(\begin{array}{ccc|ccc} 1 & 2 & 3 & 1 & 0 & 0 \\ 0 & 1 & -3 & -2 & 1 & 0 \\ 0 & 0 & -1 & -5 & 2 & 1 \end{array}\right) \sim \left(\begin{array}{ccc|ccc} 1 & 2 & 3 & 1 & 0 & 0 \\ 0 & 1 & -3 & -2 & 1 & 0 \\ 0 & 0 & \boxed{1} & 5 & -2 & -1 \end{array}\right) \qquad R_3' = (-1)R_3$$

$$\sim \left(\begin{array}{ccc|ccc} 1 & 2 & 0 & -14 & 6 & 3 \\ 0 & 1 & 0 & 13 & -5 & -3 \\ 0 & 0 & 1 & 5 & -2 & -1 \end{array}\right) \qquad \begin{array}{l} R_2' = R_2 + 3R_3 \\ R_1' = R_1 - 3R_3 \end{array}$$

$$\sim \left(\begin{array}{ccc|ccc} 1 & 0 & 0 & -40 & 16 & 9 \\ 0 & 1 & 0 & 13 & -5 & -3 \\ 0 & 0 & 1 & 5 & -2 & -1 \end{array}\right) \qquad R_1' = R_1 - 2R_2$$

$$\underbrace{\begin{pmatrix} -40 & 16 & 9 \\ 13 & -5 & -3 \\ 5 & -2 & -1 \end{pmatrix}}_{A^{-1}}$$

determinant

$$\underbrace{A_{n\times n}}_{\det} \qquad \underbrace{A_{m\times n}}_{\det \times}$$

$|A| = 2 \qquad |A^3| = ?$

$(|A|)^3 = 2^3$

$$|A^p| = |A|^p$$

(ii) $\det(AB) = \det(A)\det(B)$

(iii) $\det(A^{-1}) = \left(\det(A)\right)^{-1}$

$|A| = 2, \quad |B| = 3 \qquad \det(A^5 B A^{-1})$